PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

paediatric fractures

1. Why children's fractures are different

- **Periosteum** is thick, strong and osteogenic: it limits displacement, acts as a hinge for reduction (keep the intact periosteal side as the hinge) and produces fast, abundant callus.
- **Bone is more porous and less brittle:** it fails in compression (**torus/buckle**), on the tension side only (**greenstick**), or bends without fracture (**plastic deformation**, most often ulna and fibula; can block reduction of the partner bone).
- **Remodelling** is greatest when the child is young, the fracture is **near a fast-growing physis**, and the deformity is **in the plane of motion of the adjacent joint**. **Rotational deformity does not remodel.** Intra-articular step-off does not remodel.
- **Overgrowth:** femoral shaft fractures in children aged about 2-10 overgrow by roughly 1-2 cm, which is why 1-2 cm of shortening (overlap) is accepted in a spica.
- **Ligaments are stronger than the physis:** an apparent "sprain" in a child is often a physeal injury (e.g. Salter-Harris I of the distal fibula).
- **Fastest-growing physes:** proximal humerus (about 80% of humeral growth) and distal radius (about 75-80% of radial growth) in the upper limb; **distal femur (about 70% of femoral length)** and proximal tibia (about 55-60% of tibial length) around the knee. Roughly 1 cm a year at the distal femur and 0.6 cm at the proximal tibia.

2. The physis and Salter-Harris

- **Zones (epiphysis to metaphysis):** reserve (resting), proliferative, **hypertrophic** (maturation, degeneration, **provisional calcification**). Fractures usually propagate through the hypertrophic zone, the weakest region; the germinal (reserve and proliferative) cells stay with the epiphysis, which is why most physeal fractures grow normally.
- **Salter-Harris classification:**

Type	Fracture line	Notes
I	Through the physis only	Infants and young children; radiographs may be normal (look for widening); distal fibula, proximal humerus in neonates
II	Physis + metaphyseal fragment (**Thurston-Holland**)	**Commonest (about 75%)**; the periosteum is intact on the fragment side
III	Physis + epiphysis (intra-articular)	Needs anatomical reduction (e.g. juvenile Tillaux)
IV	Metaphysis + physis + epiphysis	Intra-articular and crosses the germinal layer; **ORIF**; high risk of a bar (e.g. medial malleolus, lateral condyle humerus)
V	Crush of the physis	Usually diagnosed retrospectively when growth arrest appears; worst prognosis
VI (Rang)	Perichondral ring injury	Peripheral bar, angular deformity (lawnmower injuries)

- **Peterson type I** (transverse metaphyseal fracture extending into the physis, distal radius) is worth knowing for FET-level questions.

- **Highest growth-disturbance risk: distal femur** (even SH I and II, because of the undulating physis), proximal tibia, distal tibia SH III/IV.
- **Physeal bar:** diagnose with **MRI (or CT) mapping**. Resect with interposition (fat, cement) if it involves **under 50%** of the physis and the child has **at least 2 years (about 2 cm) of growth left**. Otherwise complete the arrest (epiphysiodesis), deal with the other bone, and correct the angulation and length.
- **Park-Harris growth arrest lines** that converge on the physis indicate a bar; parallel lines mean symmetrical growth.

3. Elbow (the high-yield region)

Ossification centres (CRITOE)

- **C**apitellum about 1 year, **R**adial head 3-5, **I**nternal (medial) epicondyle 5-7, **T**rochlea 7-9, **O**lecranon 9-11, **E**xternal (lateral) epicondyle 11-13 (girls earlier).
- The **medial epicondyle always appears before the trochlea**: an ossified fragment in the joint before the trochlea ossifies is an incarcerated medial epicondyle.
- **Anterior humeral line** passes through the middle third of the capitellum; the **radiocapitellar line** passes through the capitellum in every view (if not, look for Monteggia).
- **Posterior fat pad sign** = effusion = occult fracture (supracondylar, lateral condyle, radial neck).

Supracondylar humerus fracture

- Commonest elbow fracture in children (peak 5-7 years), mechanism fall on the outstretched hand. **Extension type about 97%**, flexion type about 3%.
- **Gartland classification (with Wilkins modification):**

Type	Description	Treatment
I	Undisplaced; posterior fat pad only	Above-elbow cast 3 weeks
IIA	Displaced, posterior cortex intact (hinge), no rotation	Closed reduction; pin if anterior humeral line misses the capitellum
IIB	Hinged but rotated or translated	Closed reduction and pinning
III	Completely displaced	**Closed reduction and percutaneous pinning**
IV (Leitch)	Multidirectionally unstable (periosteal hinge lost)	Pinning; often diagnosed under the image intensifier

- **Nerves: anterior interosseous nerve is the commonest** (extension type), then radial, median and ulnar.
  - **Posterolateral displacement** of the distal fragment (proximal spike goes anteromedial): **median nerve and brachial artery**.
  - **Posteromedial displacement** (spike goes anterolateral): **radial nerve**.
  - **Flexion type: ulnar nerve**.
  - **Iatrogenic:** ulnar nerve with a medial pin, especially when the elbow is hyperflexed (the nerve subluxes anteriorly).
  - Most nerve palsies are neurapraxias that recover in 2-3 months.
- **Pinning: 2 or 3 divergent lateral pins** separated at the fracture site are stable for most fractures with no ulnar nerve risk. Crossed pins are biomechanically stiffer; if a medial pin is needed, make a small incision and extend the elbow before placing it.
- **Vascular status:**

  - **Pink pulseless hand** (perfused, no palpable pulse) after reduction: admit, elevate and observe closely. Explore only if perfusion deteriorates or there is **associated median nerve palsy** (the artery may be tethered or entrapped).
  - **White pulseless hand** after closed reduction: **urgent open exploration** of the brachial artery (anterior approach).
  - **Compartment syndrome** (Volkmann ischaemia) risk is higher with ipsilateral forearm fracture (floating elbow) and with **casting in more than 90 degrees of flexion**.
- **Assessing reduction: Baumann angle** (shaft axis vs capitellar physis on AP; about 70-75 degrees, compare with the other side; change over 5 degrees = coronal malalignment), anterior humeral line, **teardrop**, medial and lateral column integrity.
- **Cubitus varus ("gunstock")** is the commonest late complication. It is due to **malunion (medial column collapse, internal rotation)**, not growth disturbance. It is cosmetic but can cause posterolateral rotatory instability, ulnar nerve symptoms and snapping triceps. Treat with a **lateral closing-wedge (French) osteotomy** or dome osteotomy.

Lateral condyle fracture

- Second commonest elbow fracture (age 5-10). It is **intra-articular and physeal (behaves like SH IV)**, pulled by the extensor muscles, so it displaces and has a high **non-union** risk.
- **Milch I** (lateral to the trochlear groove, SH IV, stable elbow) vs **Milch II** (into the trochlear groove, SH II-like, elbow unstable). **Jakob/Weiss** grade by displacement: under 2 mm, 2-4 mm with intact cartilage hinge, over 4 mm and rotated.
- **Internal oblique view** shows the true displacement best. An **arthrogram or MRI** shows whether the articular cartilage

hinge is intact.

- **Treatment:** under 2 mm, cast with weekly radiographs (it can displace); **over 2 mm, ORIF** (posterior dissection is avoided: the blood supply enters posteriorly, so dissecting there causes **AVN of the condyle**).
- **Complications:** non-union, leading to **cubitus valgus** and **tardy ulnar nerve palsy** years later; lateral spur (overgrowth, cosmetic "pseudo-varus"); fishtail deformity.

Medial epicondyle fracture

- Ages 9-14; a traction (apophyseal) injury; about half are associated with **elbow dislocation**.
- **Absolute indication for surgery: incarceration of the fragment in the joint** (after dislocation reduction). Relative: ulnar nerve dysfunction, valgus instability in throwers or gymnasts, displacement in the dominant arm of an athlete.
- Most displaced fractures do well non-operatively (fibrous union is usually painless).

Radial neck, Monteggia and other forearm injuries

- **Radial neck:** angulation **under 30 degrees** is accepted; 30-60 degrees, closed reduction (Patterson manoeuvre, Israeli technique); over 60 degrees or translated, percutaneous joystick or **intramedullary (Metaizeau) reduction**. Avoid open reduction where possible: **loss of forearm rotation (stiffness) is the commonest complication**, and AVN of the radial head and radioulnar synostosis follow open reduction.
- **Monteggia (Bado):** I anterior radial head dislocation with apex-anterior ulnar fracture (**commonest in children**), II posterior, III lateral (often ulnar metaphyseal greenstick), IV anterior dislocation with both-bone fracture. In children, **restoring ulnar length and alignment usually reduces the radial head**; plastic deformation of the ulna must be corrected.

A **missed Monteggia** (chronic) needs ulnar osteotomy with annular ligament reconstruction.

- **Galeazzi:** radial shaft fracture with distal radioulnar joint disruption; the paediatric equivalent is a distal ulnar physeal separation.
- **Forearm shaft:** acceptable alignment is generous in children under about 9 (angulation up to about 15 degrees, bayonet apposition, malrotation up to about 45 degrees); in older children less (about 10 degrees angulation, 30 degrees malrotation). Mid-shaft radius malunion limits rotation most. **Elastic stable intramedullary nails** for unstable fractures in older children. **Refracture** is common (about 5%) after greenstick fractures: complete the greenstick or cast long enough.
- **Distal radius:** the commonest paediatric fracture overall. Huge remodelling capacity: in a child with 2 years of growth left, up to about 20-25 degrees of sagittal angulation is acceptable. **Do not re-manipulate a physeal (SH II) fracture after 5-7 days** (physeal damage).
- **Nursemaid's elbow (pulled elbow):** annular ligament subluxation in children aged 1-4. Reduce by supination with flexion or by hyperpronation (more successful).

4. Proximal humerus and clavicle

- Proximal humerus physeal fractures (SH II in adolescents, SH I in younger children) remodel enormously (up to about 60 degrees accepted in young children). Operate only in adolescents with severe displacement or soft-tissue interposition (biceps tendon).
- **Neonatal clavicle fracture** (difficult delivery, pseudoparalysis): heals without treatment. Differentiate from **obstetric brachial plexus palsy** and **congenital**

pseudarthrosis of the clavicle (right side, painless, no callus, subclavian artery pressure theory).

- **Medial clavicle physeal injury** (the last physis to close, about 23-25 years): posterior displacement can compress mediastinal structures. **CT**, then reduce in theatre with a thoracic surgeon available.

5. Spine in children

- **SCIWORA** (spinal cord injury without radiographic abnormality) is due to the elastic paediatric spine; **MRI** shows the lesion.
- Upper cervical injuries predominate under about 8 years (large head, horizontal facets). **Pseudosubluxation of C2 on C3** (up to about 4 mm) is normal; use the **Swischuk line**.
- Use a **back-board with an occipital recess or shoulder roll** in children under about 6 (the large occiput flexes the neck).
- **Chance (flexion-distraction) fracture** with a lap belt: look for associated **bowel injury**.

6. Pelvis and hip

- Pelvic fractures in children often need less fixation (elastic pelvis), but **associated visceral injuries** are frequent. The triradiate cartilage can arrest (shallow acetabulum).
- **Avulsion fractures in adolescent athletes:** ASIS (sartorius), **AIIS (rectus femoris)**, ischial tuberosity (**hamstrings**, the one that may need fixation if displaced over 1.5-2 cm), lesser trochanter (iliopsoas; **in an adult, a lesser trochanter avulsion = pathological fracture until proved otherwise**).
- **Delbet classification of paediatric hip fractures:**

Type	Site	Frequency	AVN risk
I	Transepiphyseal (with or without dislocation)	Rare	Highest, near 100% with dislocation
II	**Transcervical**	**Commonest (about 45-50%)**	About 50%
III	Cervicotrochanteric (basicervical)	About 30%	About 25%
IV	Intertrochanteric	About 10-15%	Lowest, about 10%

- Treat as an emergency: **urgent anatomical reduction**, stable fixation (avoid crossing the physis if possible in young children; stability is more important than sparing it), and **capsular decompression**. Complications: **AVN (commonest and most serious)**, coxa vara, non-union, premature physeal closure.

7. Femoral shaft

Age / size	Treatment
0-6 months	**Pavlik harness** or posterior splint
6 months-5 years	**Immediate (or early) hip spica**; accept up to about 2 cm shortening, about 15 degrees varus/valgus and about 20 degrees anterior/posterior angulation (less with age)
5-11 years	**Flexible (titanium) elastic nails** for length-stable fractures in children **under about 45-49 kg**; submuscular bridge plating or external fixation for length-unstable or comminuted fractures

Age / size	Treatment
Over about 11 years or heavier	**Rigid intramedullary nail through a lateral trochanteric entry**. Avoid piriformis entry: the lateral ascending cervical branches of the **medial femoral circumflex artery** pass there, so it risks **AVN of the femoral head**

- **Elastic nails:** two nails, each about **40% of the medullary canal diameter** (combined about 80%), pre-bent to the same curve so they balance. **Complications:** loss of length in unstable patterns, nail-end irritation, malunion in heavier children.
- A **femoral fracture in a child who is not yet walking** (under about 1 year) is a red flag for **non-accidental injury**.
- **Distal femoral physeal fractures** (often SH II) have a high rate of growth arrest (about 40-50%) even when well treated: warn the family and follow up for 1-2 years.

8. Knee

- **Tibial spine (eminence) fracture:** ages 8-14, the paediatric ACL equivalent. **Meyers-McKeever:** I undisplaced, II anterior hinge, III completely displaced (IIIA displaced, IIIB rotated), IV comminuted (Zaricznyj). The **intermeniscal ligament or anterior horn of the medial meniscus** can block reduction. Type I and reducible II: long-leg cast in extension. Irreducible II, III and IV: **arthroscopic or open reduction and fixation** (suture or screw).
- **Tibial tubercle avulsion:** adolescent boys jumping; Ogden classification. Watch for **compartment syndrome** (anterior tibial recurrent artery).
- **Patellar sleeve fracture:** a cartilage sleeve pulled off the inferior pole (ages 8-12); the radiograph shows only a small

fleck of bone and patella alta, so the injury is underestimated. ORIF.

- **Proximal tibial metaphyseal fracture (Cozen):** a seemingly innocent undisplaced fracture in a child aged 3-6 develops **progressive genu valgum** over the next year from asymmetric overgrowth. Warn the parents; most remodel; avoid early osteotomy.
- **Proximal tibial physeal (SH) fractures:** risk of **popliteal artery injury** (the artery is tethered at the trifurcation).

9. Tibia and ankle

- **Toddler's fracture:** undisplaced spiral fracture of the distal tibia in a child just walking (9 months-3 years). May only show on an oblique view or at 10-14 days. Usually accidental.
- **Transitional fractures** occur in adolescents, because the distal tibial physis closes over about 18 months: **central first, then medial, then lateral**.
  - **Juvenile Tillaux:** SH III avulsion of the **anterolateral epiphysis** by the **anterior inferior tibiofibular ligament**, as only the lateral physis is still open.
  - **Triplane:** SH IV in three planes. It looks like SH III on the AP view and SH II on the lateral view; two-, three- or four-part types.
  - **CT** for both. **Articular displacement over 2 mm** needs reduction and fixation (screws parallel to the joint, avoiding the physis if growth remains).
- **Distal tibial SH III/IV (medial malleolus):** high risk of physeal bar and **varus** deformity. Anatomical ORIF.

10. Non-accidental injury (NAI)

- Red flags: a history inconsistent with the injury or the child's development, delay in presentation, injuries of different ages, and **any fracture in a non-ambulant infant**.
- **High-specificity fractures: classic metaphyseal lesions** (corner or bucket-handle fractures, from shaking or twisting), **posterior rib fractures** (squeezing), scapula, spinous process and sternum fractures.
- Moderate specificity: multiple or bilateral fractures, fractures of different ages, epiphyseal separations, vertebral body fractures, digital fractures in non-ambulant children, complex skull fractures.
- **Commonest fracture in abuse overall: diaphyseal long-bone fracture** (humerus or femur in infants); **humeral shaft fracture under 3 years** is suspicious.
- **Workup:** full **skeletal survey** in children under 2 (repeat at about 2 weeks), ophthalmology (retinal haemorrhages), head imaging, bloods (bone profile, vitamin D, coagulation). Differentials: **osteogenesis imperfecta**, rickets, scurvy, Caffey disease, congenital syphilis, normal metaphyseal variants.
- The treating doctor has a duty to **report concerns** (in India: child protection services; POCSO where sexual abuse is suspected) and to admit the child for safety.

> Exam pearl: **AIN is the commonest nerve injured in a supracondylar fracture; the ulnar nerve is the commonest iatrogenic injury (medial pin).** A **lateral condyle fracture over 2 mm displaced needs ORIF**, and its late complication is **tardy ulnar palsy from cubitus valgus**.

> Exam pearl: **Delbet II is the commonest paediatric hip fracture; Delbet I has the highest AVN risk. Piriformis-entry nails are avoided in children because of AVN.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.